Predicting pathological complete response from pre-treatment breast biopsy slides with an open pathology foundation model

Oct 3, 2026 · @Someone

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Abstract

Background. Response to neoadjuvant chemotherapy in breast cancer is known only at surgery, roughly six months after diagnosis, yet it shapes adjuvant treatment and long-term prognosis. Computational models that read the diagnostic biopsy slide could move that estimate to the time of diagnosis. Published approaches depend on manual tissue delineation, networks pre-trained on natural images, or access-restricted pathology foundation models trained on hundreds of patients.

Methods. We studied 107 patients with a digitised pre-treatment core biopsy and a recorded pathological outcome from a single institution (T-CAIREM dataset). Each whole-slide image was cropped to tissue and embedded with Phikon, an openly available pathology foundation model pre-trained on 40 million H&E tiles from The Cancer Genome Atlas, yielding a 768-dimensional slide representation. Five principal components of this embedding were combined with molecular subtype, chemotherapy regimen, Nottingham grade and clinical tumour size in logistic regression models predicting pathological complete response (pCR; residual cancer burden [RCB] 0 versus RCB I–III). Discrimination was measured by the area under the receiver operating characteristic curve (AUC) [validation scheme to state]. Feature contributions were examined with SHAP.

Results. Image features alone predicted pCR with an AUC of 0.704. Clinical variables alone reached 0.856 and molecular subtype alone 0.858. Combining subtype with image features gave the highest discrimination (AUC 0.882), and clinical variables plus image features reached 0.862. In SHAP analysis, three image components ranked directly after the two most influential subtype indicators.

Conclusions. A frozen, openly available pathology foundation model, used without any manual annotation, extracts response-related signal from routine pre-treatment biopsies and adds to molecular subtype. The compact pipeline suits single-centre cohorts and is fully reproducible from public weights. External validation is required.

Keywords: breast cancer; neoadjuvant chemotherapy; pathological complete response; residual cancer burden; computational pathology; foundation model; Phikon

Introduction

Neoadjuvant chemotherapy is standard for locally advanced and high-risk early breast cancer, and its effect is judged only at surgery. Patients with locally advanced disease typically receive about 24 weeks (eight cycles) of chemotherapy before resection, after which pathological response is graded on the specimen, most often as pathological complete response (pCR) or by the residual cancer burden (RCB) index.

That grade matters to patients beyond the operating room. In a pooled analysis of 5161 patients from 12 cohorts, RCB was prognostic for long-term survival regardless of pre-treatment clinicopathological features and of hormone receptor and HER2 subtype [1]. At the trial level pCR is a weak surrogate for long-term outcome, but it remains the accepted way to estimate an individual patient's residual risk and to select candidates for further adjuvant therapy [2]. Residual disease after neoadjuvant treatment now triggers escalation: trastuzumab emtansine for residual HER2-positive disease [3] and capecitabine for residual HER2-negative disease [4].

A reliable estimate of response at the time of diagnosis would let clinicians and patients plan with that information months earlier. Tissue biomarkers already carry part of the signal. Tumour-infiltrating lymphocytes predict response in triple-negative and HER2-positive disease [5], a low tumour–stroma ratio in the pre-treatment biopsy is associated with higher pCR rates in HER2-negative disease [6], and the Ki-67 labelling index is associated with pCR [7]. Each of these is a single, hand-defined feature scored by a pathologist or a dedicated algorithm.

Computational pathology offers a way to read the whole biopsy slide at once. Systematic reviews report that deep-learning models can discriminate pCR well in retrospective cohorts, but heterogeneous methods and scarce external validation limit their clinical use [8, 9]. The published H&E-based models fall into four broad approaches: segmentation of pre-defined tissue features [10–14], convolutional networks pre-trained on natural images [15–18], small self-supervised or hybrid encoders [19, 20], and, most recently, large pathology foundation models combined with multiple-instance learning [21–23]. All three foundation-model studies used the UNI family of encoders, which require an access request and were trained on cohorts of 174 to 950 patients.

Two practical gaps remain. Many centres hold only one or two hundred annotated neoadjuvant cases, too few to train attention-based models with many parameters. And reproducibility is constrained when the encoder is gated or was pre-trained on non-public data. Open pathology foundation models pre-trained solely on public data, such as Phikon [24], have not, to our knowledge, been evaluated for neoadjuvant response prediction.

We therefore asked whether a frozen, openly available pathology foundation model,

We therefore asked whether a frozen, openly available pathology foundation model, applied to the routine pre-treatment biopsy without any manual annotation, can predict pCR in a single-centre cohort, and whether its image features add to molecular subtype and routine clinical variables.

Methods

Study population

We used the T-CAIREM breast cancer digital pathology dataset [institution, accrual years and ethics approval to add]. Eligible patients had locally advanced breast cancer treated with neoadjuvant chemotherapy followed by surgery, a digitised pre-treatment core needle biopsy stained with haematoxylin and eosin (H&E), and a recorded pathological outcome. A total of 107 patients met these criteria [exclusions and patient flow to add]. Each patient contributed one pre-treatment whole-slide image (WSI).

Outcome

The primary outcome was pCR, defined as RCB 0 on the surgical specimen. Patients with RCB I, II or III were classed as not achieving pCR [number of pCR events to add].

Clinical variables

Four routinely available variables were used: molecular subtype (triple-negative, HER2-positive, luminal A, luminal B), chemotherapy regimen, Nottingham histological grade, and clinical tumour size (cT, mm). After encoding, these formed seven model inputs [encoding of categorical variables and handling of missing values to confirm].

Slide processing and feature extraction

Each WSI was cropped to the tissue region, excluding background [tissue-detection method, magnification, tile size and artefact quality control to state]. Image features were extracted with Phikon, a ViT-Base vision transformer (86 million parameters) pre-trained by self-supervision (iBOT) on 40 million pan-cancer tiles from The Cancer Genome Atlas [24]. Phikon was used frozen, with no fine-tuning on study data, and produced a 768-dimensional representation per slide [tile-to-slide aggregation, e.g. mean pooling, to state].

Phikon was chosen because its weights are openly downloadable without an access request, it was pre-trained exclusively on public data, and its compact embedding suits a small cohort. To limit overfitting, the 768-dimensional slide embeddings were reduced by principal component analysis (PCA) to five components [whether PCA was fitted within training folds to confirm].

Model development

We fitted logistic regression models with five input sets: (i) image components only; (ii) clinical variables only; (iii) molecular subtype only; (iv) clinical variables plus image components; and (v) molecular subtype plus image components. The fullest model therefore had at most 12 inputs for 107 patients. Logistic regression was preferred over attention-based multiple-instance learning because it has few parameters and its coefficients can be inspected directly.

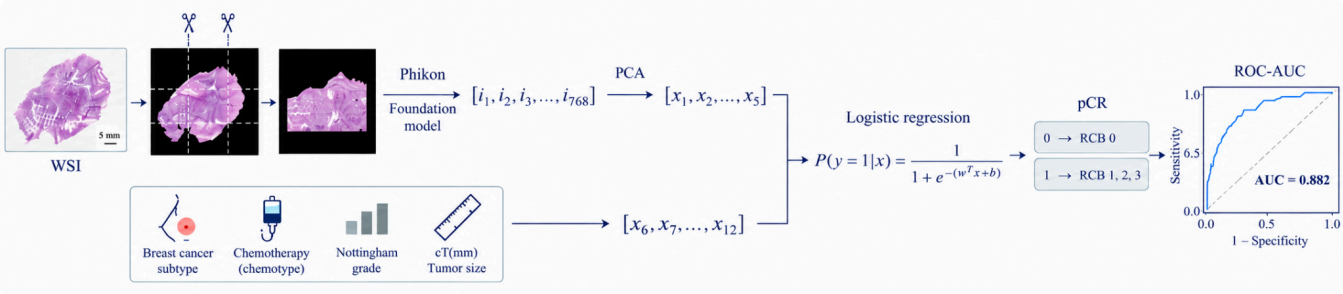


Figure 1. Study pipeline. The pre-treatment whole-slide image (WSI) is cropped to tissue and embedded by the Phikon foundation model into 768 features, reduced by PCA to five components (x1–x5). These are combined with clinical variables (x6–x12: subtype, chemotherapy regimen, Nottingham grade, clinical tumour size) in a logistic regression predicting pCR (RCB 0 vs RCB I–III).

Evaluation

Discrimination was measured by the area under the receiver operating characteristic curve (AUC) [validation scheme, e.g. stratified k-fold cross-validation with repeats, and confidence-interval method to state]. All models were evaluated on the same patients.

Interpretability

Feature contributions were estimated with SHAP (Shapley additive explanations) [27] for the subtype-plus-image model, giving the direction and size of each feature's effect on the predicted probability of pCR for every patient.

Literature comparison

To place the approach in context, we reviewed 14 published studies that predict neoadjuvant response from H&E slides [10–23]. For each we extracted the cohort, endpoint, image representation and reported AUC from the full text, and grouped the studies by how they read the slide.

Reporting

The study is reported following TRIPOD+AI [28] [checklist to complete].

Results

Cohort

The cohort comprised 107 patients across all four molecular subtypes [Table 1 with age, subtype, grade, tumour size, regimen and pCR rate to add].

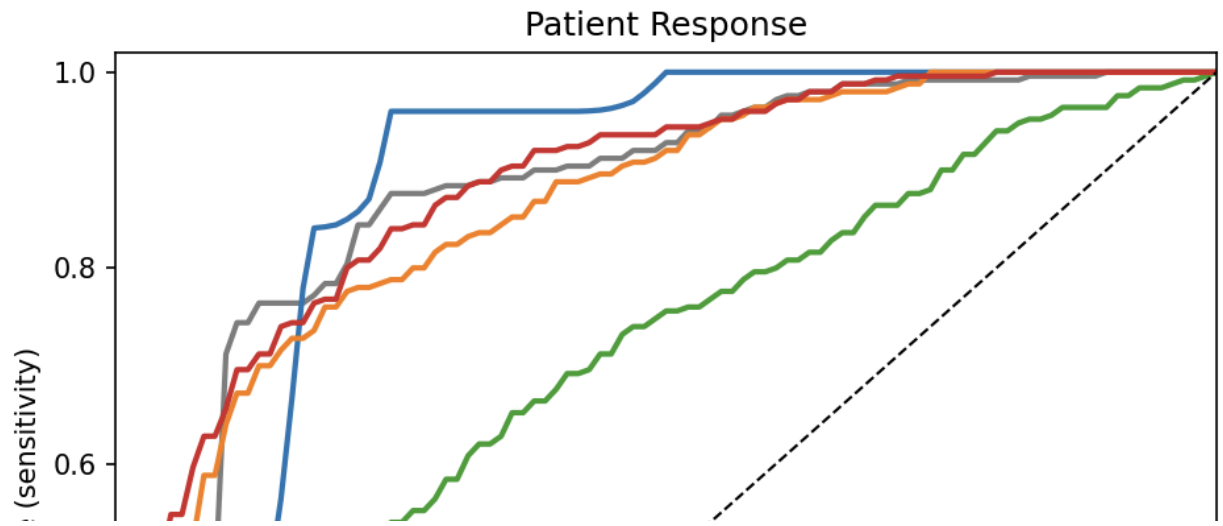
Model performance

Combining molecular subtype with image features gave the best discrimination, with an AUC of 0.882 (Table 2, Figure 2). Image features alone reached 0.704, showing that the biopsy slide carries response-related signal on its own. Adding image features to subtype raised the AUC from 0.858 to 0.882; adding them to the full clinical set raised it from 0.856 to 0.862.

Table 2. Discrimination of pCR by model inputs (n = 107).

Model inputs	Inputs (n)	AUC	Change from adding images
Image components only	5	0.704	—
Clinical variables only	7	0.856	—
Molecular subtype only	[4]	0.858	—
Clinical variables + image	12	0.862	+0.006
Molecular subtype + image	[9]	**0.882**	+0.024

[95% confidence intervals and a paired DeLong test for each image-added model to add.]



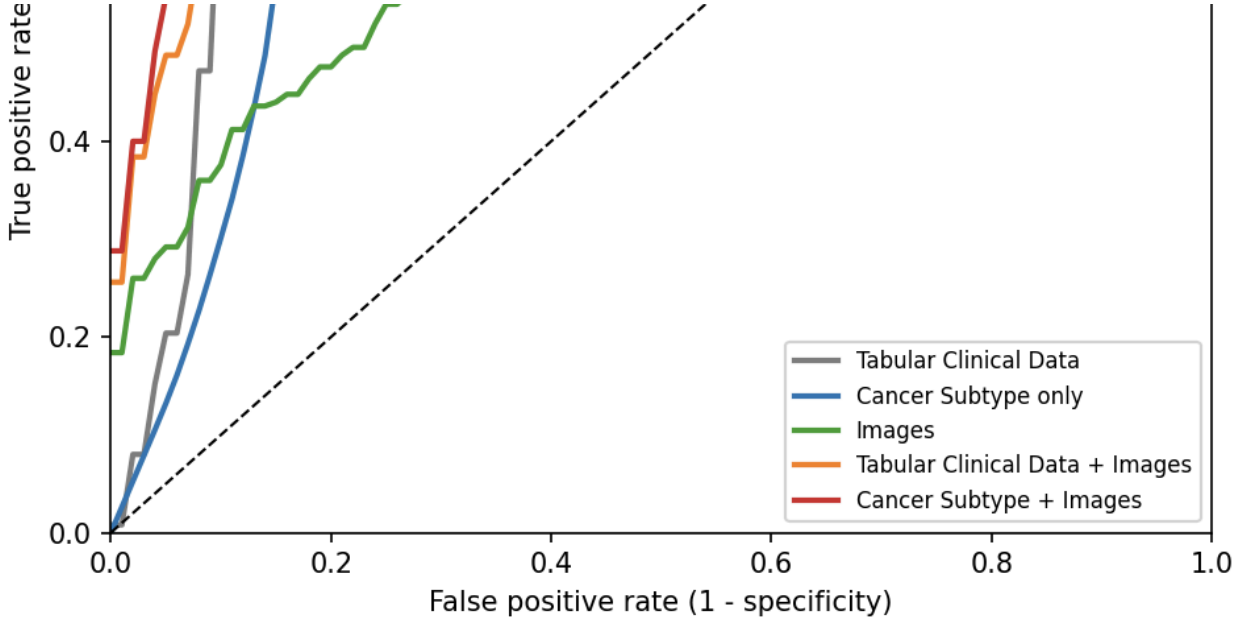
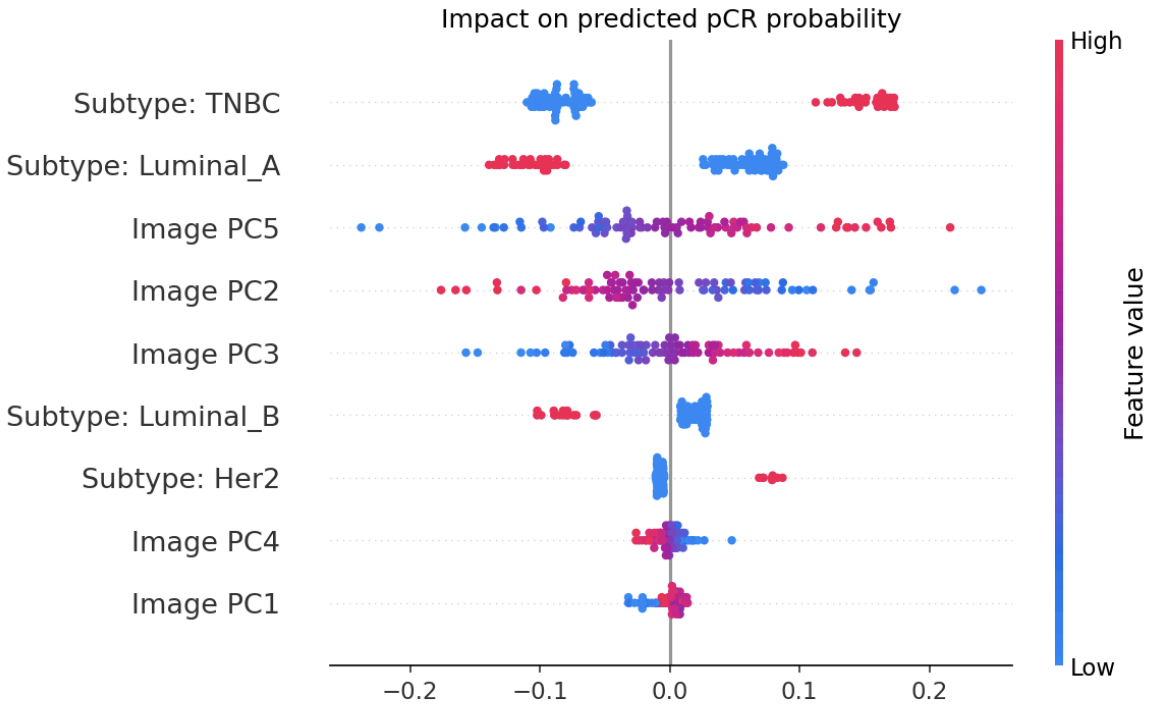


Figure 2. Receiver operating characteristic curves for the five models on the same 107 patients. The dashed line marks chance.

Feature contributions

In the subtype-plus-image model, the two most influential features were subtype indicators: triple-negative subtype raised the predicted probability of pCR and luminal A lowered it, consistent with clinical experience (Figure 3). Three image components (PC5, PC2 and PC3) ranked next, above the luminal B and HER2 indicators. High values of PC5 and PC3 pushed predictions toward pCR, and high values of PC2 pushed them away. PC4 and PC1 contributed little.



SHAP value (impact on model output)

Figure 3. SHAP summary plot for the subtype-plus-image model. Each dot is one patient; position shows the feature's effect on the predicted probability of pCR and colour shows the feature value.

Discussion

Principal findings

A frozen, openly available pathology foundation model, applied to the routine pre-treatment biopsy without any annotation, extracted features that predicted pCR on their own (AUC 0.704) and added to molecular subtype (0.858 to 0.882). The image signal was not a restatement of subtype: three image components ranked directly after the two strongest subtype indicators in SHAP analysis. The whole pipeline used one public encoder, five image components and a logistic regression fitted to 107 patients.

Comparison with prior approaches

Published H&E-based models of neoadjuvant response fall into four groups (Table 3). Reported AUCs are not directly comparable across studies, because cohorts, subtypes, endpoints and validation designs differ; they are given for context only.

Table 3. Prior H&E-based approaches to predicting neoadjuvant response, and how this study differs.

Approach	Studies	What it requires	Reported AUC	This study
A. Feature identification	Dodington 2021 [10], Saednia 2022 [11], Huang 2023 [12], Aswolinskiy 2023 [13], Cui 2026 [14]	Segmentation of TILs, stroma, tumour bed or nuclei before measuring pre-defined features; IMPRESS also used paired IHC	0.59–0.91	No manual delineation: the whole biopsy is embedded automatically
B. ImageNet-pretrained CNNs	Li 2021 [15], Li 2022 [16], Zeng 2024 [17], Sturm 2025 [18]	Networks pre-trained on natural photographs, fine-tuned or used as feature extractors	0.70–0.85 (image alone)	A pathology foundation model pre-trained on 40 million H&E tiles, used frozen

		on 205 to 874 patients		
C. Small self-supervised or hybrid encoders	Valderrama 2026 [19], Lin 2026 [20]	A small ViT (384 features) with narrow pre-training, fused with an ImageNet CNN, on 500 to 1,543 patients	0.65–0.81 external	One open, pan-cancer encoder used alone
D. Foundation model with MIL	Khan 2025 [21], Hou 2026 [22], Zhou 2026 [23]	UNI or UNI-2 (gated access, CC BY-NC-ND licence) with attention-based MIL on 174 to 950 patients	0.63–0.82 external	First use of Phikon; public-data encoder; 12-input model suited to a single-centre cohort

Against feature identification. Methods in group A measure what investigators choose in advance, and each segmentation step adds annotation effort and error. Their performance also varied by subtype: IMPRESS reached 0.90 in HER2-positive disease but 0.59 in triple-negative disease on external validation [12]. Our pipeline requires no delineation, so it can capture stromal and immune context that no one pre-specified.

Against ImageNet-pretrained networks. Group B models transfer features learned from photographs of everyday objects to tissue. Alone, they reported AUCs of 0.70 to 0.85, often on several hundred patients, and several gained most from added clinical data [16, 17]. Our image-only model reached 0.704 with 107 patients and no network training, because the encoder already represents histology.

Against small self-supervised and hybrid encoders. Group C combined a small ViT, pre-trained on one cancer type in one case [20], with an ImageNet CNN. Two-branch designs make it hard to know which component carries the signal. We used a single larger encoder pre-trained across cancer types, so the source of the image signal is unambiguous. One study in this group reported near-perfect AUCs (0.994–0.999) [20], which warrant independent replication.

Against foundation models with MIL. Group D is methodologically closest. All three studies used the UNI family [21–23], which requires an access request and was pre-trained partly on non-public institutional slides [25]. To our knowledge this is the first evaluation of Phikon for neoadjuvant response. Phikon is downloadable without restriction and pre-trained only on public data, so the image features can be reproduced exactly by any group. Its compact embedding, reduced here to five components, keeps the final model to at most 12 inputs; attention-based MIL fits far more parameters, which is a risk with a cohort of this size.

Choice of encoder

Phikon scores below newer encoders on public tissue-classification benchmarks. On three breast benchmarks (BACH, BRACS, BreaKHis) its mean balanced accuracy is 0.679, against 0.813 for UNI-2 and 0.795 for Midnight-12k [26]. None of these benchmarks measures response to therapy, so the size of any gain for this task is unknown. That Phikon nonetheless captured response-related signal suggests the finding reflects the tissue rather than encoder strength, and stronger encoders may raise performance further. A direct comparison on this cohort is planned.

Clinical implications

The model uses only material that exists for every patient at diagnosis: one H&E biopsy slide and routine clinical variables. An estimate of pCR at that point could inform discussions about the expected course of treatment and identify patients likely to need post-surgical escalation, months before surgery confirms it. Any such use requires external and prospective validation first.

Limitations

- The cohort is small (n = 107) and from a single institution, and no external validation was performed.
- The gain from adding image features to subtype (+0.024) is modest; its confidence interval has not yet been reported and may include zero.
- Molecular subtype dominated prediction, as expected, and the incremental value of the slide over a richer clinical model was small (+0.006).
- PCA is unsupervised, so the five retained components were not selected for their relation to response, and response-relevant variation may lie in discarded components.
- Chemotherapy regimens were not uniform, and the binary pCR endpoint discards the gradation captured by the full RCB index.
- Phikon is not the strongest available encoder on public benchmarks.

Future work

The next steps are validation in an external cohort, confidence intervals and paired tests for the image contribution, a direct comparison with UNI-2 and Midnight-12k using the same pipeline, and prediction of the full RCB class.

Conclusion

An openly available pathology foundation model, used frozen on routine pre-treatment biopsies with no manual annotation, predicted pathological complete response to neoadjuvant chemotherapy and added to molecular subtype (AUC 0.882). The pipeline is compact enough for single-centre cohorts and reproducible from public weights. Validation in independent cohorts is the next step.

Declarations

- **Ethics approval:** [to add]
- **Data availability:** T-CAIREM dataset [access terms to add]. Phikon weights are publicly available at huggingface.co/owkin/phikon.
- **Code availability:** [repository to add]
- **Funding:** [to add]
- **Competing interests:** [to add]
- **Acknowledgements:** 2026 National Health Datathon organisers; T-CAIREM [to confirm]. Pipeline figure acquisition image: T-CAIREM, Matt Bielecki.

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